

Technology Stack

Date	9 july 2026
Team ID	StudyAI-Smart AI educate Companion&Quiz Generator
Project Name	
Maximum Marks	2 Marks

Define Your Technology Stack:

This document defines the technology stack used to build StudyAI - a React + Vite frontend for the student interface, a Python Flask backend handling authentication and file processing, Firebase Firestore for data storage, and the Groq API (Llama 3.3 70B) for generating AI-powered summaries, flashcards, and quizzes -chosen to keep the app fast, scalable, and easy to maintain.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	React.js + Vite, Tailwind CSS, React Router, Axios, Chart.js, React Dropzone	Fast build tooling, utility-first responsive styling, easy API calls, and drag-and-drop file uploads

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python Flask, Flask-JWT-Extended, Flask-CORS	Lightweight REST API framework with fast development and secure token-based auth
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	Firebase Firestore	NoSQL, real-time, handles unstructured document/quiz data with easy scaling
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	Vercel (frontend), Render (backend), Gunicorn (WSGI server)	Free-tier friendly, easy CI-based deployment, production-ready server for Flask
5	Version Control & CI/CD	GitHub	Seamless collaboration, version history, and integration with Vercel/Render auto-deploy

	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	Groq API (Llama 3.3 70B Versatile), Firebase Authentication	Enables fast AI-generated summaries/flashcards/quizzes and secure user login